

Department of Electrical and Computer Engineering
Virginia Polytechnic Institute and State University

ECE 2804
Integrated Design Project
Spring 2026

Course Policies

Class Meetings: A scheduled 2-hour weekly lab plus a seminar from 2:30 – 3:30pm on Fridays (Torgersen 2150).

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Office Hours: By instructor.

Overview: The main part of this course involves working with a partner to solve an open-ended design problem. During the first week of the course, you will complete a lab to reacquaint yourself with your measurement equipment and the design of some basic circuits. Following this you will choose a project to work on, be assigned a partner, and answer a list of questions about topics specific to your project. This will constitute Milestone 1. After that you will work on your project for the rest of the course. We will make every effort to avoid sections with an odd number of students, but if the issue can't be resolved, we may require a student to work by themselves on a version of the project with fewer design objectives.

You are required to attend the entire 2-hour lab session each week to work with your partner and meet with your instructor. There is also a Friday seminar, which you are required to attend, that will cover topics such as professionalism, technical writing, presentation skills, ethics, and talks by industry representatives and alumni touching on careers, teamwork, ethics, and leadership.

The project you will be involved in may be significantly different from those you have done in previous courses. It is open-ended, meaning there are many possible solutions, and we are not going to lead you through it step-by-step. It is your job to figure out the steps, justifying your decisions with knowledge from your previous courses and new information you find on your own, and put it all together into a working, well-documented project. You will provide weekly updates to your instructor, and your instructor will discuss your progress with you and provide a grade and feedback on your design choices and tests.

Materials:	Basic electronics equipment (USB oscilloscope, proto-board, digital multimeter) and a parts kit. The scope will be lent to you for the semester on the first day of class. The other material you should have obtained in ECE 1004. Any specialized parts associated with your project will be provided. Since projects and labs in other courses may require proto-board space, you should ensure that you have multiple protoboards available.
Use of Instructional Materials:	Students may record or take notes on class sessions, laboratories, office hours, or other instructional activities for their personal use only. All course materials—including slides, recordings, assignments, and other instructional content—are protected under copyright law and may not be shared, posted, or distributed without either the instructor's or university's express written consent. Unauthorized dissemination of these materials, including modified, edited, or excerpted portions, is a violation of university expectations and may be addressed under the Code of Student Conduct and/or the Honor System.
Prerequisites:	1004 and 2024 (ability to use an oscilloscope and DMM, design and build standard op-amp circuits, analyze the operation of first and second order circuits, design a first order filter, measure the frequency response of a circuit, design a 555-based oscillator, use diodes and BJTs in basic circuits, design rectifier circuits, program in C using the

Arduino IDE, use an Arduino to read analog voltages and generate PWM waveforms)
2514 (system design process and additional programming experience)
2544 (digital hardware, digital design, and finite state machines)

Objectives:

After successfully completing this course, students will be able to:

1. Solve open-ended design problems (i.e. problems with many possible solutions), involving both analog and digital hardware, based largely on material in the prerequisite courses, but requiring the ability to learn new material as needed.
2. Document their solution to an open-ended design problem in a well-written technical report that clearly explains the problem, their solution, and validates that their solution actually solves the problem.
3. Present their solution to an open-ended design problem in a well-designed and effectively-delivered presentation that clearly explains the problem, their solution, and validates that their solution actually solves the problem

Academic
Accommodations:

Virginia Tech welcomes students with disabilities into the University's educational programs. The University promotes efforts to provide equal access and a culture of inclusion without altering the essential elements of coursework. If you anticipate or experience academic barriers that may be due to disability, including but not limited to ADHD, chronic or temporary medical conditions, deaf or hard of hearing, learning disability, mental health, or vision impairment, please contact the Services for Students with Disabilities (SSD) office (540-231-3788, ssd@vt.edu, or visit ssd.vt.edu).

If you require accommodations (such as Services for Students with Disabilities or religious holiday conflicts), please let your instructor know during the first two weeks of the semester. Accommodations for extended illness and for personal or family emergencies should be handled by the Dean of Students staff.

Honor System:

The Undergraduate Honor Code pledge that each member of the university community agrees to abide by states:

"As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do."

Students enrolled in this course are responsible for abiding by the Honor Code. A student who has doubts about how the Honor Code applies to any assignment is responsible for obtaining specific guidance from the course instructor before submitting the assignment for evaluation. Students are strongly discouraged from misusing sites such as Chegg and CourseHero, as well as misusing ChatGPT and other Generative Artificial Intelligence. Students are strongly encouraged to consult their faculty members regarding the use of such outside materials as the misuse of these sources may constitute a violation of the Honor Code. Ignorance of the rules does not exclude any member of the University community from the requirements and expectations of the Honor Code.

For additional information about the Honor Code, please visit:
<https://www.honorsystem.vt.edu/>

In this course, you are encouraged to discuss the course material with the instructor, Undergraduate Learning Assistants, and, in general terms, with other students. All submitted work, however, *must be your own and reflect your understanding*.

Permissible AI: In this course, use of generative AI tools is permitted for all assignments, given that the student is using the AI tool to contribute to their learning of the course content, but not to replace original work. You must properly acknowledge when and how AI tools have been used in your submissions. Failure to disclose AI use or submitting AI-generated work as entirely your own without critical engagement will constitute a violation of the Honor Code. If you are ever uncertain about the use of AI, it is your responsibility to ask your faculty member before proceeding.

Weekly Update: Starting with week 4, each student will submit a weekly update presentation. The Weekly Update presentation is due by 11:59pm the day before your lab meeting. These presentations will form the basis for discussions with your instructor during the weekly lab meetings, where each student is required to attend and present their work. The instructor will provide feedback and grade your progress and documentation. The grade will be based on the quality of your designs and analysis, quantitative evidence of performance, degree of progress, and ability to discuss your designs and progress. The number

of slides used in your presentation is up to you, but each student will only have 5 minutes to present their work. Before the end of class, each student will upload a comment to their Canvas submission including notes taken during their discussion and what their action items are for the next week.

Attendance during each lab session is required, but exceptions will be handled on a case-by-case basis.

Milestone Reports: After two Weekly Updates, each group will then submit a Milestone Report the following week documenting their work on a given milestone. These Milestone Reports will help form a baseline for the design verification section of the Final Report. Guidelines for the various reports are appended to this syllabus.

Teamwork Expectations: For all Milestone Reports, Presentations, and the Final Report, it is expected that both group members contribute approximately equal amounts of work to the project. If your instructor believes that the work is not being split evenly when viewing your submissions or talking about project progress during lab, they reserve the right to assign different grades on team assignments. For Weekly Updates, each student will be graded individually on their work/presentation.

Friday Seminar: The common-time Friday seminar will be used to provide information on professionalism and career topics from speakers within and outside Virginia Tech. Attendance is **required**. The material in the seminars is important course content and, therefore, if you need to miss a seminar, you must let your instructor know through email before the seminar, and then after the recording is posted to Canvas, you must watch the video and submit a 500-word essay that clearly and thoughtfully discusses the main points, their significance to you individually, and your reflection on how you'll use them moving forward.

If your instructor believes you missed a significant amount of any given seminar, they may ask you to write the essay afterwards. If the essay is not submitted or is inadequate, a one-step grade penalty will be assessed to your final course grade. The essay is due before the next Friday seminar and should be submitted to your instructor via email. It is an honor code violation to use AI to produce these essays. The option of writing an essay is to provide you with some flexibility

regarding your schedule but is limited to a **maximum of three seminars** to promote in-person engagement.

Final Report: Each group will submit a final report, the format of which is appended below. The report should be divided into as many subsections as necessary and attribution to the author of each section provided on the authorship page. The report should be written so that the reader can understand how to redesign the final product with different specifications if asked to. The complete code listing should be in the appendix and not in the body of the report.

Presentations: Each student group will give an oral presentation of their work at weeks 7 and 14 to document their project at the midpoint and end of their work. The presentations should be no longer than 8 minutes in length, exclusive of questions and answers. The target audience is engineers at your experience level that are not familiar with your project. The midpoint presentation will introduce the project, discuss your overall approach, and discuss some of your design, testing, and validation results to date. The final presentation will introduce the project, discuss your overall approach, and present results that validate the correct operation of the overall system. There will be questions at the end of each presentation and the presentations will be graded by the instructor.

Awards: The most successful and innovative groups may be given the opportunity to present a poster at the senior Major Design Experience (MDE) poster session at the end of the following semester and have their names listed in the MDE brochure as being “Best in Course.” You will have the opportunity to interact with industry representatives, which might lead to internship or other opportunities.

Grading Policy:	Introductory Lab	10%
	Weekly Updates	25% total
	Milestone 1 Report	5%
	Milestone 2 Report	6%
	Milestone 3 Report	12%
	Milestone 4 Report	12%
	Final Report	10%
	Presentation 1	5%
	Presentation 2	5%
	Seminar Attendance	10%

Notes:

1. While the above policy outlines the grading percentage breakdown for each part of the course, points are assigned based on the quality of the work. A rough indication of how we view quality is given by the following scale:

A – Showing an **exceptional** level of creativity, design skill, understanding, analysis, or project enhancements

B – Deliverable works and is understood

C – Deliverable partially works, or works but is poorly understood

D – Deliverable does not work or why project works is not understood

F – Essentially no significant work

2. The instructor reserves the right to reduce the grade of any late submission or not accept late submissions.
3. While some of the submitted work is a group effort and will typically receive a group grade, it is possible for individuals in a group to receive different grades if it is clear that one member's contribution differs significantly from another member's. Penalties will be assessed individually.
4. Each group member should build and test their own final project. Although each member may have designed only part of the final project, each member is responsible for understanding and being able to answer detailed questions about the entire project, so it is critical that you completely understand your partner's work.
5. Exceptions will be considered on a case-by-case basis.

Schedule:

Week*	Task	Description
1	Introductory Lab Part 1, Pick project	Do the first part of a formal lab exercise to review your prerequisite knowledge.
2	Introductory Lab Part 2	Do the second part of a formal lab exercise to review your prerequisite knowledge. Begin working with your partner on Milestone 1.
3	Milestone 1	Research your project and provide answers to the questions posed in the project description for Milestone 1.
6	Milestone 2	Score based on meeting milestone design and testing goals, clear description of all relevant schematics, test results that demonstrate your subsystems work correctly, clear descriptions of your code, and a clear description of the reasoning behind all design choices in your hardware and software.
7	Presentation 1	Presentation to discuss progress of project so far and next steps in project design.
10	Milestone 3	Same as Milestone 2.
13	Milestone 4	Same Milestone 2 with entire system complete and working.
14	Final Report Final Presentation	Report and presentation to discuss the final design of the project and its capabilities in meeting the design requirements.

*The end of your 2-hour lab session is considered the end of the week. Thus, the end of your first lab session is considered the end of week 1. You will be assigned partners prior to the beginning of your second lab session.

Deadlines:

The official deadlines for each assignment will be the deadlines posted on Canvas.

Assignment	Deadlines
Seminar Attendance	Start of seminar each Friday (see Canvas for list of Friday seminar dates)
Introductory Lab, Part 1	Week 2 two days before lab
Introductory Lab, Part 2	Week 2 two days after lab
Weekly Updates	Weeks 4, 5, 8, 9, 11, 12 one day before lab
Milestone Reports	Weeks 3, 6, 10, 13 two days before lab
Presentation 1	Week 7 before lab
Presentation 2	Week 14 before lab
Final Report	Week 14 before lab

Submission Guidelines

Weekly Update:

1. Title should be of the form Week X Update: XXXX Project, followed by the name of the authors
2. For each milestone task that you worked on (no more than one task per slide), provide:
 - a. Clear and concise description of what was achieved (in bullet form)
 - b. Designs with detailed schematics, calculations, flowcharts, or code snippets.
Note: the variable names in schematics and code must match the names in your calculations and descriptions.
 - c. Simulations, measurements, or other **quantitative** evidence that shows you are meeting the performance specifications you have chosen.
 - d. Note: If both teammates worked together on a task, they must decide who will present it during their Weekly Update. Both of them should not present the same material or at the same time.
3. On the final slides, briefly discuss:
 - a. Your plan for the next week, including tasks to work on and goals you set
 - b. Any technical problems that **are preventing** progress and your plans to resolve the problems
 - c. Any immediate questions you'd like to discuss during your meeting
4. Note: To help stay within the time limit, consider putting only the most important test results and figures in the presentation, and consider including additional slides after the end of your presentation that could be helpful when discussing with your instructor

Milestone Report (Maximum of 4 pages in single-column format using 1.15 line spacing and 11-point Calibri font):

1. Title should be of the form Milestone X Report: XXXX Project, followed by the names of the authors.
2. For each required deliverable, provide:
 - a. Reasoning behind all of your design choices. Include schematics, calculations, and key code snippets.
 - b. Test results that **quantitatively** show you have met the specified milestone objective and performance specifications you have chosen.
 - c. Note: To help meet the page limit, consider putting only the most important test results and figures in the body of the report, and include any additional test results and figures in an appendix at the end of the report. Include a complete code listing and complete schematics.
3. On an additional page at the end of the Milestone Report (which will not count against your page limit), include an Authorship Table that states which team members worked on which sections of the report, including a breakdown for who worked on the technical work of each section and who wrote each section.

Final Report Requirements (Maximum of 16 pages in single-column format using 1.15 line spacing and 11-point font):

Section	Excellent (A)
1. Introduction	The goals of the project are clearly introduced and any goals beyond the minimum are noted.
2. High Level Design	A block diagram of the entire system is introduced. The choices that resulted in this solution are clearly discussed. Reasoning should be supported by credible evidence, and a high level of critical thinking should be evident. The concept of operation of the overall system is clearly described.
3. Detailed Design & Verification	At the beginning of this section, a schematic of your entire system is provided. The design of each individual subsystem is described in detail with evidence of superior design choices. The details of calculations behind all choices (component values, sampling rates, frequencies, etc.) are provided and clearly explained. Detailed schematics and verification measurements of each subsystem are provided and discussed. The clarity should be such that after reading a subsystem description, an engineer should have no trouble designing a similar subsystem with somewhat different specification values. Non-working subsystems will be graded as 0.
4. Validation of the Overall Project	Provide and discuss measurements that show the overall project achieves the goals for the project and meets the specifications that the team designed for. Any shortcomings are clearly noted and explained. This is an end-to-end evaluation that shows the integrated system works as required. If deliverables are not met, the grade for this portion may be reduced.
5. Conclusion	A reflection on what was learned by doing the project and what they would do differently if they had to do this project again. A high level of critical thinking should be evident.
Authorship Page	A page should be attached to the report that explains which team member wrote each section and note if the other team member revised the section before submission. A substantial writing contribution should be evident for each team member.
Overall Writing	Writing should be clear, concise, and free of errors in mechanics (grammar, spelling, punctuation, sentence structure). Word usage is varied and appropriate. All figures and tables should be appropriately captioned and clearly readable.

Rubrics:

These general rules must be followed in all submitted work. If not, points in addition to those in the rubrics can be deducted. These guidelines can be found on the course Canvas page.

- a. The General Guidelines for Lab Submissions must be followed for all assignments submitted for this course.
 - b. The Wiring Guidelines must be followed for all physical circuits built and tested.
1. Milestone 1: 5 points (note: good answers must tie your research to the project goals)
 - a. 0 – nothing turned in
 - b. 1 – not all questions answered
 - c. 2 – all questions answered
 - d. 3 – Some good answers to the questions
 - e. 4 – Good answers to all questions
 - f. 5 – Some excellent answers that show exceptional research or thought
2. Weekly Updates: 5 points
 - a. 1 point for giving an update with some details on progress made
 - b. 1 point for updating with discussion of progress, concerns, issues, future goals
 - c. 1 point for documenting adequate progress toward completing the milestone tasks (~20%/week for each team member)
 - d. 1 point for including sufficient details of the design and testing process, problems encountered, and upcoming goals
 - e. 0.5 points for writing and uploading notes and action items based on in-class discussion with instructor to Canvas as a comment on the submission by end of class
 - f. Up to 0.5 point for discussing excellent exploration of designs and design choices
3. Milestone Reports 2-4: 10 points
 - a. Meeting milestone objectives: 3 points
 - i. 0 – no objectives met
 - ii. 1 – some objectives met
 - iii. 2 – half of the objectives met
 - iv. 3 – all objectives met
 - b. Test results (quantitatively if possible) that clearly backup claims in (a): 2 points
 - i. 0 – test results do not support objectives
 - ii. 1 – test results support most objectives
 - iii. 2 – test results support all objectives, quantitatively if possible
 - c. Clear engineering reasoning behind all design choices: 2 points
 - i. 0 – no reasons given or trial and error used
 - ii. 1 – clear (quantitative if possible) reasons for most choices
 - iii. 2 – clear (quantitative if possible) reasons for all choices
 - d. Up to 1 point for discussing excellent aspects of the project in meeting (or exceeding) the design requirements
 - e. Clarity of writing, figures, and organization: 2 points
 - i. 0 – unclear
 - ii. 1 – fair, could use improvement

- iii. 2 – good, no improvement needed
- 4. Final Report: 10 points
 - a. Introduction 0.5 point
 - i. 0 – poor description
 - ii. 0.5 – goals of the project clearly articulated
 - b. High Level Design: 1 point
 - i. 0 – No block diagram or description of the system unclear in parts
 - ii. 1 – Block diagram, clear description of system operation, well-reasoned choices for system components.
 - c. Detailed Design & Verification of Subsystems: 3 points
 - i. 1.5 points for clearly reasoned design choices and clear calculations.
 - ii. 1.5 points for detailed schematics and validation **measurements** for each subsystem
 - d. Validation of the Overall Project: 2 points
 - i. Up to 2 points for tests that show the fully operational system meets the design requirements
 - e. Conclusion: 0.5 point
 - i. 0 – fluff
 - ii. 0.5 – a short, critical reflection on what was learned and what they would do differently
 - f. Authorship. If not balanced between the partners, use differential grading.
 - g. Overall Writing: 3 points
 - i. 0 – figures, tables, or writing unclear in most places
 - ii. 1 – figures, tables, or writing unclear in many places
 - iii. 2 – all figures and tables have good captions and are clearly readable. Writing is clear in most places.
 - iv. 3 – all figures and tables have good captions and are clearly readable. Writing is clear in all places.
- 5. Presentations: 5 points
 - a. Delivery: 2 points
 - i. 1 point for smooth delivery (talking equally, confident, clear, engaging, minimal reliance on notes/slides)
 - ii. 0.5 points for staying within time limit
 - iii. 0.5 points for excellent delivery
 - b. Organization and Clarity: 2 points
 - i. 1 point for easily readable slides that flow in a logical order and follow guidelines presented in the seminar about effective presentations
 - ii. 0.5 points for good use of figures
 - iii. 0.5 points for excellent slides that enhance the impact of the presentation
 - c. Technical Details: 1 point
 - i. 1 point for technical details that are thoughtfully chosen and well conveyed to the intended audience